



Core Concepts

Metrics

Available Metrics

Retrieval Augmented Generation

Response Relevancy

[Copy page](#)

Answer Relevancy

The **Answer Relevancy** metric measures how relevant a response is to the user input. It ranges from 0 to 1, with higher scores indicating better alignment with the user input.

An answer is considered relevant if it directly and appropriately addresses the original question. This metric focuses on how well the answer matches the intent of the question, without evaluating factual accuracy. It penalizes answers that are incomplete or include unnecessary details.

This metric is calculated using the `user_input` and the `response` as follows:

1. Generate a set of artificial questions (default is 3) based on the response. These questions are designed to reflect the content of the response.
2. Compute the cosine similarity between the embedding of the user input (E_o) and the embedding of each generated question (E_{g_i}).
3. Take the average of these cosine similarity scores to get the **Answer Relevancy**:

$$\text{Answer Relevancy} = \frac{1}{N} \sum_{i=1}^N \text{cosine similarity}(E_{g_i}, E_o)$$

$$\text{Answer Relevancy} = \frac{1}{N} \sum_{i=1}^N \frac{E_{g_i} \cdot E_o}{\|E_{g_i}\| \|E_o\|}$$

[latest](#)

Where:

- E_{g_i} : Embedding of the i^{th} generated question.
- E_o : Embedding of the user input.
- N : Number of generated questions (default is 3, configurable via `strictness` parameter).

Note: While the score usually falls between 0 and 1, it is not guaranteed due to cosine similarity's mathematical range of -1 to 1.

Example

```
from openai import AsyncOpenAI
from ragas.llms import llm_factory
from ragas.embeddings.base import embedding_factory
from ragas.metrics.collections import AnswerRelevancy

# Setup LLM and embeddings
client = AsyncOpenAI()
llm = llm_factory("gpt-4o-mini", client=client)
embeddings = embedding_factory("openai", model="text-embedding-3-small", client=client)

# Create metric
scorer = AnswerRelevancy(llm=llm, embeddings=embeddings)

# Evaluate
result = await scorer.ascore(
    user_input="When was the first super bowl?",
    response="The first superbowl was held on Jan 15, 1967"
)
print(f"Answer Relevancy Score: {result.value}")
```

Output:

  latest ▼

```
Answer Relevancy Score: 0.9165088378587264
```

Synchronous Usage

If you prefer synchronous code, you can use the `.score()` method instead of `.ascore()`:

```
result = scorer.score(  
    user_input="When was the first super bowl?",  
    response="The first superbowl was held on Jan 15, 1967"  
)
```



How It's Calculated

Example

Question: Where is France and what is it's capital?

Low relevance answer: France is in western Europe.

High relevance answer: France is in western Europe and Paris is its capital.

To calculate the relevance of the answer to the given question, we follow two steps:

- **Step 1:** Reverse-engineer 'n' variants of the question from the generated answer using a Large Language Model (LLM). For instance, for the first answer, the LLM might generate the following possible questions:
 - *Question 1:* "In which part of Europe is France located?"
 - *Question 2:* "What is the geographical location of France within Europe?"
 - *Question 3:* "Can you identify the region of Europe where France is situated?"

  latest ▼

- **Step 2:** Calculate the mean cosine similarity between the generated questions and the actual question.

The underlying concept is that if the answer correctly addresses the question, it is highly probable that the original question can be reconstructed solely from the answer.

Legacy Metrics API

The following examples use the legacy metrics API pattern. For new projects, we recommend using the collections-based API shown above.

Deprecation Timeline

This API will be deprecated in version 0.4 and removed in version 1.0. Please migrate to the collections-based API shown above.

Example with SingleTurnSample

```
from ragas import SingleTurnSample
from ragas.metrics import ResponseRelevancy

sample = SingleTurnSample(
    user_input="When was the first super bowl?",
    response="The first superbowl was held on Jan 15, 1967",
    retrieved_contexts=[
        "The First AFL-NFL World Championship Game was an American football game played on January 15, 1967, at the Los Angeles Memorial Coliseum in Los Angeles."
    ]
)

scorer = ResponseRelevancy(llm=evaluator_llm, embeddings=evaluator_embeddings)
await scorer.single_turn_ascore(sample)
```


 latest ▼

Output:

0.9165088378587264



December 11, 2025

December 11, 2025

GitHub